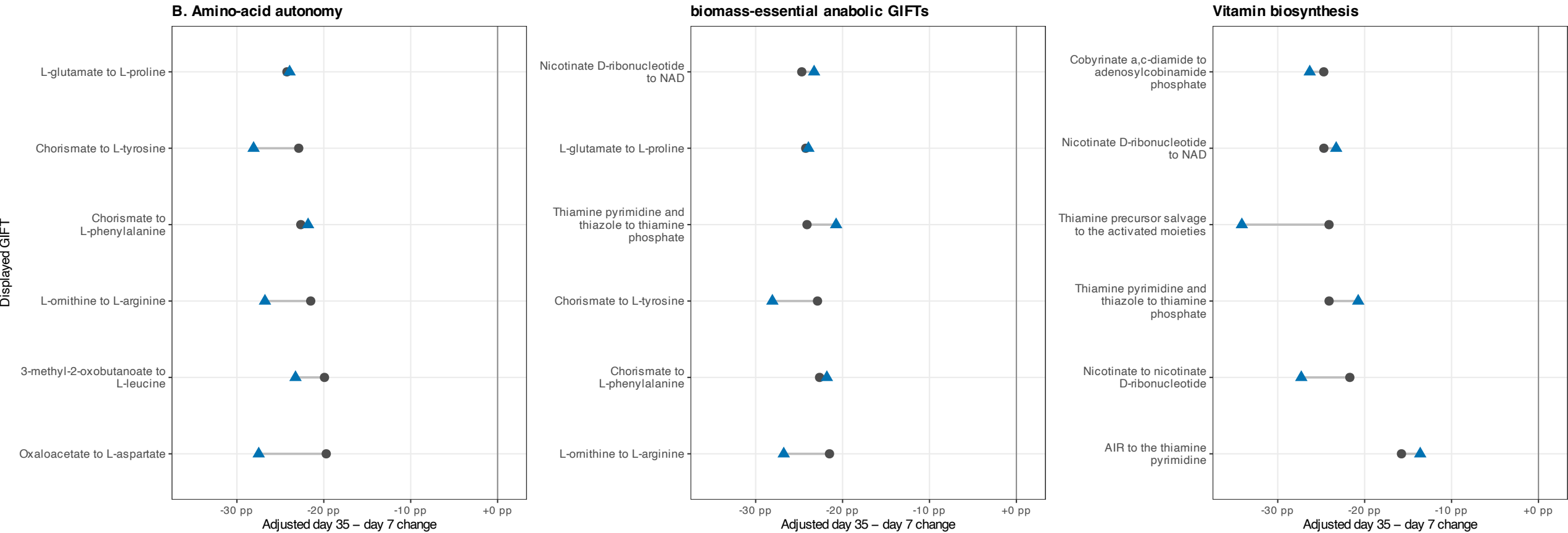
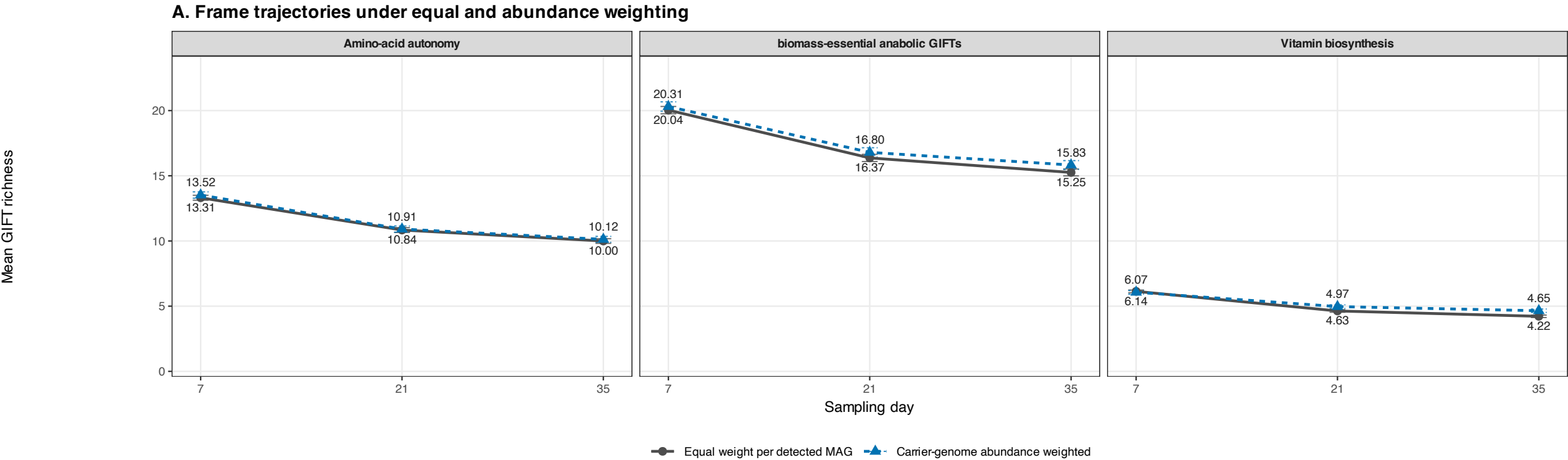


Abundance weighting preserves the principal frame-level declines

Five of six focal contrasts remain beyond the ± 1 -GIFT margin; vitamin biosynthesis at day 21 remains negative but its magnitude is uncertain



Abundance is closed within each sample's detected MAGs. Carrier abundance is encoded capability distribution, not activity, expression or flux.